

# Companion LXXXVIII: Emulator-Accelerated RDCC Inference Using Neural Transport Surrogates and Multi-Fidelity Simulation Flows

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## Abstract

Inference in the Relaxation-Driven Cyclic Cosmology (RDCC) requires repeated evaluation of summary statistics across a range of  $\alpha$  values. Full weak-lensing simulations are computationally expensive, making direct likelihood evaluation impractical. In this work, we introduce emulator-accelerated RDCC inference using neural transport surrogates and multi-fidelity simulation flows. Neural transport surrogates emulate the distribution of persistence diagrams, Minkowski functionals, and shear power spectra as a function of  $\alpha$ , while multi-fidelity flows combine low- and high-resolution simulations into a unified generative model. This framework reduces simulation cost by orders of magnitude and enables fast, accurate, and scalable RDCC inference for Euclid-like surveys.

## 1 Introduction

Inference in RDCC requires evaluating summary statistics for many values of the relaxation parameter  $\alpha$ . Full weak-lensing simulations are expensive, and the summary statistics used in previous Companions—transport distances, Minkowski functionals, and shear power spectra—are sensitive to non-linear structure formation.

To make RDCC inference scalable, we introduce:

- neural transport surrogates for topological statistics,
- multi-fidelity simulation flows combining low- and high-resolution data,
- emulator-based likelihood acceleration.

This reduces computational cost while preserving accuracy.

## 2 Neural Transport Surrogates

We construct a surrogate model  $S_\theta$  that emulates the distribution of persistence diagrams as a function of  $\alpha$ :

$$D \sim S_\theta(\alpha).$$

The surrogate is trained using:

- neural Schrödinger flows (Companion LXXX),

- unified diffusion–OT geometry (LXXXI),
- transport-based sensitivity maps (LXXXII).

The surrogate predicts:

- persistence-diagram distributions,
- transport distances  $\mathcal{T}_\ell(\alpha)$ ,
- their derivatives  $\partial\mathcal{T}_\ell/\partial\alpha$ .

### 3 Multi-Fidelity Simulation Flows

We combine:

- low-resolution simulations (cheap, noisy),
- high-resolution simulations (expensive, accurate),

into a single flow-based generative model:

$$f_{\text{MF}} = \lambda f_{\text{low}} + (1 - \lambda) f_{\text{high}}.$$

Flow-matching ensures consistency across fidelities:

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[ \|f_{\text{MF}}(x) - f_{\text{high}}(x)\|^2 \right].$$

This yields high-resolution accuracy at low-resolution cost.

### 4 Emulator-Accelerated Likelihood Evaluation

Given the neural joint likelihood  $\mathcal{L}_\theta$  from Companion LXXXVI, we replace expensive simulation calls with surrogate predictions:

$$\mathcal{L}_{\text{emul}}(\mathbf{s} \mid \alpha) = \mathcal{L}_\theta(\mathbf{s} \mid S_\theta(\alpha)).$$

This accelerates:

- posterior sampling,
- Fisher forecasts,
- sensitivity analyses,
- calibration loops (Companion LXXXVII).

### 5 Accuracy Diagnostics

We evaluate emulator accuracy using:

- score discrepancies  $\Delta s$ ,
- transport-based distances between surrogate and true diagrams,
- posterior-predictive checks,
- coverage tests for  $\alpha$ .

These diagnostics ensure that the emulator preserves RDCC signatures.

## 6 Conclusion

Emulator-accelerated inference is essential for applying RDCC to Euclid-like surveys. Neural transport surrogates emulate topological statistics, multi-fidelity flows combine simulations of different resolutions, and emulator-based likelihood evaluation accelerates inference by orders of magnitude. This Companion establishes the computational backbone for scalable, accurate RDCC parameter estimation.

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